Recap: QQ Plots and adjustments

To explore the fit of a (theoretical distribution) to observed data:

- Produce a QQ plot
  - If theoretical distribution is **N(0, 1)**, use `qqnorm(…)`
  - For other distributions use `qqplot(Q,Qe,…)` or `plot(Q,Qe,…)` with appropriate theoretical quantiles Q and empirical quantiles Q_e.

- We can add y=x line on plot [$Q_e(p_i) = Q(p_i)$] using `abline(0,1)`

- If points, on the graph, do not (approximately) lie on $Q_e(p_i)=Q(p_i)$ (or y=x) line, the fit of the theoretical distribution is not good.

- To determine a line that fits points on the plot:
  1) A line that passes through 1st and 3rd quartiles, use `qqline(…)`
     E.g. for a N(0,1) plot, the intercept and slope of the fitted line can be computed using the 1st and the 3rd quartiles points obtained from `qnorm(c(0.25,0.75))` and `quantile(data,c(0.25,0.75))`

  2) A regression line that is resistant to outliers, use `lqs(Qe~Q)`
     If we wish to set the intercept equal to 0, then use `lqs(Qe~Q-1)`

- If fitted line is $Q_e(p_i) = a + bQ(p_i)$, we can determine if this "transformation" corresponds to a known distribution (will not always be the case).